

COURSE DESCRIPTION			
Program	Artificial Intelligence, Artificial Intelligence & Machine Learning, Biomedical Engineering, Cloud Computing and Big Data, Cyber Forensics and Information Security, Computer Science and Technology, Computer Science & Engineering, Computer Engineering, Computer Science and Engineering (Artificial Intelligence & Machine Learning), Electronics and Communication, Electronics Engineering, Electronics Engineering (Embedded Systems), Electronics and Computer Engineering, Integrated Circuit Design & Fabrication, Instrumentation Engineering, Information Technology, Micro Electronics, Automation and Robotics, Robotic Process Automation		
Course Title	Engineering Workshop Practice		
Course Code	2009B	Semester	I & II
Type of Course	Lab	Contact Hours	45 + 45
Teaching Scheme (L: T:P:R)	0:0:3:0	Credits	1.5
CIE Marks	60	ESE Marks	40

Introduction

Workshop Practice is a basic engineering course. The knowledge of basic workshops like wood working, fitting, welding, sheet metal and electrical wiring is essential for technicians to perform his/her duties in industries. Students are able to perform various operations using hand tools, equipment and machineries in various shops and in daily life. It develops the attitude of team based work and safety awareness. This course provides an industrial environment in the educational institution.

Course Objectives

1	To familiarize safety precautions practiced in the workplace.
2	To prepare drawings of models for fabrication.
3	To identify and practice various measuring, marking, holding, striking, cutting tools and precision instruments.
4	To practice electrical wiring and soldering.
5	To familiarise with basic electronic components.
6	To operate machines, power tools, and equipment safely.

Course Pre-requisite

Topic	Course code	Course Title	Semester
Trigonometry and other basic mathematical principles			Secondary Education
Principles of electrical conductivity and basics of electronics			Secondary school
Geometric constructions and projections	1004	Engineering Drawing with CAD	I

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Identify the importance of industrial safety and hands-on fire extinguisher practice and familiarise 5S and make use of various tools and devices used in carpentry and familiarise various wood alternative materials (MDF, Multi wood, plywood, etc.) to make simple boxes and wooden shapes for making wooden panelled music systems, wooden paneled robotic arms, LED light panel, etc.	Apply
CO2	Make use of various tools, machines, instruments and power tools used in the fitting shop to make clamp/ bracket for power supply cabin, metal bracket for camera mount, metal heat sink, etc. and practice on making holes and threads in metals and taking measurements by precision instruments.	Apply
CO3	Make use of various tools, machines, instruments and power tools used in the Welding shop to make welding joints for power supply casings, server racks, switch gear panels, etc.	Apply
CO4	Make use of various tools and measuring instruments to fabricate sheet metal models for casings of amplifiers, inverters, ups, power supply units, metal enclosure for arduino project, etc.	Apply
CO5	Make use of various tools and accessories to practice electrical wiring, motor connection and perform soldering of basic electronic circuits.	Apply

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11

CO1	3	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-
CO4	3	-	-	-	-	-	-	-	-	-	-
CO5	3	-	-	-	-	-	-	-	-	-	-
Total	15	-	-	-	-	-	-	-	-	-	-
Correlation Level	3	-	-	-	-	-	-	-	-	-	-

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total	
0	0	3	0	1.5	45	1.5				60	40	100	100

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Detailed Syllabus

Expt No.	Experiment name	Practical Outcome	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
	Industrial Safety: 1. Importance of safety in industry. 2. Precautions to be observed in hazardous situations. 3. Personal Safety 4. Types of fire, class of fire, method of fire extinguishing, fire extinguishers used for different types and classes of fire. 5. Hands on Practice on Fire Extinguishers	Students acquire knowledge about the process flow to be followed for a safe working environment, skill to operate fire extinguishers.	CO1	Understand	
Module II					
1	Demonstrate the use of various wood working tools/equipment and power tools.	Get awareness of various tools and equipment related to wood working	CO1	Understand	3
2	1. Demonstrate various alternate materials of wood such as MDF (medium density fibre) board, WPC (Recycled plastic wood), plywood, etc. 2. Demonstrate various wood working processes like Marking, Planning, Cutting, Chiseling, Grooving.	Get awareness about types of wood and related materials, wood working operations.	CO1	Understand	3
3	1. Practice on cutting and drilling wooden panels using power tools. 2. Make a simple box using wooden panels , fix hinges, handle, etc	Get skill to make wooden boxes.	CO1	Apply	12
Module III					
4	Demonstrate various tools, machines, instruments and power tools used in the Fitting shop.	Get awareness of various tools and equipment related to fitting work.	CO2	Understand	3
5	Demonstrate various operations like Measuring, Marking, Cutting, and Filing.	Get awareness about different operations in fitting shops.	CO2	Understand	3
6	1. Do necessary measuring, marking, cutting, filing for making a heat sink from an aluminium piece, fabrication of clamps, brackets etc 2. Practice on drilling and screwing to fix the heat sink, clamps, brackets etc 3.) Practice on taking measurements using precision measuring instruments such as vernier calipers, micrometers, etc.	Get skill to make heat sink, clamps etc. Drilling, screwing, fixing the components in a panel.	CO2	Apply	12
Module IV					

7	1. Explain welding safety precautions. 2. Demonstrate various tools/ equipment and personal protecting equipment(PPEs) used in welding shops.	Students acquire knowledge about the safety practices to be followed specially in welding shops. Get awareness of various tools and equipment related to welding work.	CO3	Understand	3
8	1. Practice Straight line welding and Spot welding (resistance welding) 2. Joining of simple cabinets by spot welding.	Get skill on arc welding and resistance welding and knowledge on metal joining techniques.	CO3	Apply	15
Module V					
9	Demonstrate various tools and equipment used in sheet metal shops.	Get awareness of various tools and equipment related to sheet metal work.	CO4	Understand	3
10	1. Demonstrate Sheet Metal operations like sheet cutting, bending and edging. 2. Practice on simple joints	Get awareness about different operations in sheet metal work shops and on simple joints.	CO4	Apply	6
11	1. Demonstrate the use of general fasteners like screws, rivets, etc. 2. Construct sheet metal models such as power supply casings, inverter casings, UPS casings, etc.	Get skill to make different sheet metal casings , fastening techniques etc.	CO4	Apply	9
Module VI					
12	1. Explain electrical safety precautions. 2. Demonstrate various tools and accessories required for Electrical wiring and soldering. 3. Explain wiring materials generally used.	Students acquire knowledge about the safety practices to be followed specially in electrical workshops. Get awareness of various tools and equipment related to electrical work. Get knowledge about wiring materials.	CO5	Understand	3
13	1. Practice on electrical wiring like one lamp controlled by one switch and one lamp controlled by two switches(stair case wiring). 2. Make a connection of a single phase induction motor with a DOL starter.	Get skill on doing basic wiring and connecting the motor to the main supply.	CO5	Apply	9
14	1. Identification of electronic components (Passive and Active components) 2. Rectifiers and filter circuits (Half wave, full wave or Bridge Rectifiers with capacitor filter circuits) 3. Familiarisation of Arduino UNO board with basic sensor circuit. 4. Practice on soldering of simple electronic circuits.	Get awareness of various electronic components, basic electronic circuits and practice on soldering.	CO5	Apply	6

Open-ended experiment

Objective:

To encourage creativity, innovation, and application of knowledge by allowing students to design and perform an experiment beyond the prescribed list, using the concepts learned during the course.

Description:

At the end of the laboratory course, each student (or group of students) shall undertake one open-ended experiment. In this activity, students are expected to conceive, design, and implement an experiment of their own choice related to the course outcomes. The aim is to apply theoretical knowledge and laboratory skills to a new or practical problem not explicitly covered in the regular experiments.

Students may choose to:

Modify or extend an existing experiment to explore additional parameters or behaviors.
Integrate multiple concepts from the syllabus into a combined application.
Design a mini-setup or prototype to demonstrate a principle or solve a simple engineering problem.
Perform a comparative study or performance analysis of a circuit/system/component

Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		

1	Introduction to carpentry tools Study activities: • Identify basic carpentry tools: • Measuring tools: steel rule, try square, measuring tape • Marking tools: pencil, marking gauge • Cutting tools: hand saw, tenon saw • Learn the names, uses, and safety precautions of each tool. Record work: • Write a brief description of at least 8 carpentry tools. • Draw neat sketches of any 4 tools with labels.	1.5
2	Carpentry holding and striking devices Study activities: • Study holding devices: bench vice, g-clamp, c-clamp • Study striking tools: wooden mallet, claw hammer • Understand their role in making joints. Record work: • Tabulate tools vs. purpose. • Sketch one holding device and one striking tool.	1.5
3	Basic carpentry joints – theory Study activities: • Understand where each joint is used. Record work: • Draw any three carpentry joints. • Write short notes on their applications.	1.5
4	Wood and wood alternative materials Study activities: • Study natural wood (teak, pine, plywood). • Study wood alternatives: MDF (medium density fibre board), multiwood, particle board • Compare properties such as strength, cost, and applications. Record work: • Prepare a comparison table between wood and wood alternatives. • Write short notes on mdf and multiwood.	1.5
5	Practical application of tools and materials Study activities: • Observe or recall workshop practice of making joints. • Identify tools used for each joint. • Study common defects and precautions in carpentry work. Record work: • List tools required for making one joint. • Write a step-by-step procedure for making a simple joint.	1.5
6	Revision and record completion Study activities: • Revise all tools, joints, and materials. • Practice identifying tools and materials from images or samples. Record work: • Complete all drawings neatly. • Write a summary of learning outcomes. • Get the record verified	1.5
Module II		
7	Introduction to fitting shop tools Study activities: • Study basic fitting tools: • Measuring tools: steel rule, vernier caliper, micrometer • Marking tools: scribe, divider, centre punch • Learn uses, handling methods, and safety precautions. Record work: • Write short notes on any 6 fitting tools. • Draw neat sketches of 4 tools with labels	1.5
8	Holding and striking tools Study activities: • Study holding devices: bench vice, hand vice, v-block • Study striking tools: ball peen hammer, mallet • Understand their importance in joint preparation. Record work: • Prepare a table showing tools and applications. • Sketch one holding tool and one striking tool	1.5

9	Cutting and finishing tools Study activities: • Study cutting tools: hacksaw, chisels • Study finishing tools: files (flat, round, half-round) • Learn types of filing and sawing operations. Record work: • Write the procedure for hacksaw cutting. • Draw and label three types of files.	1.5
10	Machines, instruments, and power tools Study activities: • Study machines used in fitting shop: bench grinder, drilling machine • Study instruments: try square, surface gauge • Study power tools: hand drilling machine, angle grinder • Learn safety rules while using machines. Record work: • List machines and power tools with their uses. • Write safety precautions for any two machines	1.5
11	Fitting joints and practical application Study activities: • Study common fitting joints: L fitting, square gap fitting, v-joint • Identify tools and machines used for each joint. • Understand the step-by-step joint making process. Record work: • Draw neat sketches of any two fitting joints. • Write the procedure for making one fitting joint	1.5
12	Revision and record completion Study activities: • Revise tools, machines, instruments, and fit. • Practice identifying tools and joint types. • Review safety practices. Record work: • Complete all drawings and notes. write a brief conclusion of learning outcomes • Prepare the record book for submission and verification.	1.5
Module III		
13	Understand the safety practice in welding shop Study activities: • Identify safety tools used in welding shops. Record work: • Write short notes on each tools • Draw labeled sketches of safety tools.	1.5
14	Introduction to welding tools Study activities: • Identify basic welding tools: • Measuring tools: steel rule, measuring tape, try square • Marking tools: scribe, centre punch, soapstone • Hand tools: hammer, chipping hammer, wire brush • Understand the uses and handling of each tool. Record work: • Write short notes on at least 6 tools. • Draw labeled sketches of 4 tools.	1.5
15	Welding machines and power tools Study activities: • Study welding machines: arc welding machine (smaw) • Study power tools: angle grinder, cutting machine, bench grinder • Learn safety rules while handling machines and power tools. Record work: • Prepare a table showing machines, power tools, and their uses. • Write safety precautions for two machines.	1.5
16	Types of welding joints - theory Study activities: • Study common welding joints: butt joint, lap joint, corner joint, t-joint • Learn where each joint is applied. Record work: • Draw all four welding joints. write applications of each joint.	1.5
17	Practical application Study activities: • Observe or recall workshop practice of making welding joints. • Identify tools, machines, and instruments used for each joint. • Learn common mistakes and precautions during welding. Record work: • List tools and machines required for making one joint. • Write a step-by-step procedure for making a butt joint.	1.5

18	Revision and record completion Study activities: • Revise all tools, machines, instruments, and joints. • Practice identifying welding tools and joints. • Review safety practices. Record work: • Complete all drawings and notes neatly. • Write a summary of learning outcomes. • Prepare the record for submission for verification	1.5
Module IV		
19	Introduction & measuring tools Study activities: • Safety rules in sheet metal shop • Identification of sheet metal materials • Measuring instruments: steel rule, try square, divider, vernier caliper • Simple marking practice on sheet Record work: • Safety rules, measuring tools list with sketches	1.5
20	Marking & cutting tools Study activities: • Marking tools: scribe, center punch, cutting tools: straight snips, curved snips • Practice straight and curved cutting Record work: • Marking and cutting tools, cutting procedure and precautions	1.5
21	Bending & folding operations Study activities: • Bending tools: mallet, folding bars • Operations: right-angle bending, hemming Record work: • Bending operations with diagrams, hemming steps	1.5
22	Seams & joints Study activities: • Types of seams: lap seam, single seam, grooved seam • Practice simple joints Record work: • Types of seams with sketches, joint-making procedure	1.5
23	Fabrication of sheet metal model Study activities: • Pattern development • Fabrication of: tray / box / funnel (any one) • Accuracy checking using measuring tools Record work: • Model drawing, fabrication steps, tools used	1.5
24	Finishing & inspection Study activities: • Filing and edge finishing, dimensional checking, revision of tools and operations Record work: • Inspection report, learning outcomes, tool summary	1.5
Module V		
25	Understand the safety practice in electrical work shop Study activities: • Understand the safety protocols during electrical work. • Identify the instruments used in connection with safety protocols Record work: • Write the safety precautions to be followed in the electrical workshop.	1.5
26	Hand tools, instruments and materials Study activities: • Understand the hand tools and instruments used. • Understand the materials used for wiring Record work: • Draw tools used, write short notes of tools	1.5
27	Estimation for wiring Study activities: • Understand the preparation of required materials for electrical wiring. Record work: • Draw the layout diagram, circuit diagram and write the estimation, procedure of wiring exercise .	1.5

28	Connection of motor with DOL starter Study activities: • Understand the circuit diagram of connection of motor with DOL starter. Record work: • Draw the circuit diagram and procedure.	1.5
29	Identification of electronic components Study activities: • Identify electronic components (Passive and Active components), familiarise rectifiers and filter circuits. Record work: • Write the related notes in the record book.	1.5
30	Soldering practice Study activities: • Identify the tools required for soldering. • Recall the portions studied in the electrical section. Record work: • Write the procedure for soldering practice. • Finish the record.	1.5

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Workshop Technology	S.K.Hajara Chaudhary,	Median Promoters and publishers, New Delhi 2015.
2	Electrical Installation Estimation and Costing	J B Gupta	Kataria, S. K., & Sons

REFERENCE			
Sl. No.	Title	Author	Publication
1	Workshop Technology	B.S. Raghuvanshi	Dhanpat Rai and sons, New Delhi 2014
2	Work Practice Manual	K. Venkat Reddy	BS Publications, Hyderabad 2014

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	https://bharatskills.gov.in/Home/StudyMaterial?var=wyt2HdOKOUuq3xZOoZ+r+w==&Default=No	I
2	https://bharatskills.gov.in/Home/StudyMaterial?var=AzLv1odBjstKXkfVvHx0JQ==&Default=YES	II
3	https://bharatskills.gov.in/Home/StudyMaterial?var=HlryMME3UqmQNYriQLEEFa==&Default=YES	III
4	https://bharatskills.gov.in/Home/StudyMaterial?var=vNuH8QjfGrhOzGTlvfa6cA==&Default=YES	IV
5	https://bharatskills.gov.in/Home/StudyMaterial?var=WIQIPeNQdz11QpdFC+0tkQ==&Default=YES	V
6	https://bharatskills.gov.in/Home/StudyMaterial?var=WSdYV6aWadK8jUuNKxoBWg==&Default=YES	VI

Major Equipment/Instruments/Components required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
1	Bench Vice	6 "	30
2	Bench Vice	4"	15
3	Surface Plate	20" x 20"	1
4	Bench drilling Machine	1 HP	1
5	Vernier Height Gauge	300mm	1
6	Vernier Caliper	150mm	3
7	Arc welding machine	250A	1
8	Spot welding machine (Resistant welding)		1
9	Bench/Pedestal Grinder	8"	1
10	Anvil	25kg	2

11	Chop saw	2200w	2
12	Hand Shear	9"	1
13	Stakes	Standard size	5
14	Metal jack plane	8"	15
15	Carpenters Vice	8"	15
16	Hand Wood Cutting Machine (circular saw)	1200W	1
17	Electric Hand wood planer	600W	1
18	Digital Multimeter		1
19	Single phase Induction motor	1 kw	1
20	DOL Starter	415 V, 3 phase/ single phase	1
21	Angle grinder	4inch, 800 W	2
22	Rivet gun		4
23	Sheet metal bending machine	45CM	1

Major Software required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
	NIL		

Assessment Methodology - Lab

Mode	Attendance	CE	CA1	CA2	OEE	ESE
		Continuous evaluation	Practical test (first half of experiments)	Practical test (second half of experiments)	Open-ended experiment	Practical
Duration			2 hours	2 hours		3 hours
Exam mark		50	40	40	50	40
Converted to		30	15	15	10	
Marks	5	30	15		10	40
Total marks	60					40
Tentative schedule	End of semester	Every lab slot	7th week	14th week	15th week	

Scheme of Evaluation - Practical (Continuous Internal Evaluation)

Sl. No.	Criteria	Description	Marks
1	Preparation & Punctuality	Attendance and timely submission of records.	5
2	Documentation	Drawing of required figure/diagram for doing the exercise and writing the procedure.	5
3	Timely submission	Time of completion of work with respect to allowable time	5
4	Fit and finish	Fit and overall finish of models / finish of wiring model/ soldering/ estimation of wiring consumables	10
5	Submission of model	Completion and submission of exercises.	25
Total			50

Scheme of Evaluation - Practical Test & End Semester Evaluation

Category	Things to evaluate	Marks
Certified record	Completion & neatness of practical record. Proper certification by faculty.	4
Viva Voce	Understanding of exercise related theoretical knowledge.	8
Write-up / Procedure (before exercise)	Aim, circuit diagram, stepwise procedure, etc.	4
Work done	Following safety protocols, timely completion of work, Handling of tools	24
Total		40

Scheme of Evaluation - Open-ended experiment

Category	Things to evaluate	Marks
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Originality and Relevance of the Idea	Assesses how creative, innovative, and relevant the chosen experiment is to the course objectives.	10
Clarity of Objectives and Experimental Plan	Evaluates how clearly the student defines the aim, scope, and systematic plan of the experiment.	10
Correctness of Execution and Data Recording	Checks the accuracy, safety, and neatness in performing the experiment and recording observations.	10
Analysis, Interpretation, and Presentation of Results	Measures the student's ability to analyze data, interpret outcomes, and present results logically.	10
Teamwork and Innovation Demonstrated	Judges the level of collaboration, initiative, and creativity shown during the experiment.	10
Total		50

Note :

Module 1 should be taken by the respective faculty (workshop superintendent/ workshop foremen).

It can be included in the induction programme of first year students also.

Printed rough record should be given to students with blank papers provided for writing exercises.

Fair record should be prepared by the students which is duly certified and submitted at the time of end semester examination, which is mandatory.

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